



University
of Glasgow

Tuesday 10 May 2022
09:30-11:30 BST
Duration: 2 hours
Additional time: 30 minutes
Timed exam – fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

Cyber Security Fundamentals M
COMPSCI 5063

(Answer All 3 Questions)

**This examination paper is an open book, online
assessment and is worth a total of 60 marks.**

1. You are working as a police officer and are conducting an investigation into a cyber-crime.

- (a) One of the pieces of evidence gathered is a notepad which you are going through. On the last page there is a note that stands out as it does not make any sense. You start thinking that it might be referring to ciphers and decide to attempt to break them. (In order for you to get full marks you will need to demonstrate the whole procedure and mention which ciphers are being used).

[15 marks]

7 marks for every correct conversion and 1 mark for clear explanation and answer to the questions posed.

Note: 1st: 2 → (to the right)

2nd: passwd (keywd) → ZCGNFJXQX

- (b) On a different page of the notepad you find the following: DB Password = ASECRETMESSAGE. You try ASECRETMESSAGE as a password, but it does not work. There is a chance this is a clue to derive the password through substitution of characters. Please state at least 3 different passwords deriving from this information and in comparison, with password setup guidelines discussed in the course defend the level of strength that these passwords you think provide. Your answer should not exceed 250 words.

[5 marks]

2 marks for the list of three potential passwords and 3 marks for discussing different factors that affect the strength of a password alongside with a comparison to your findings.

[Total: 20 marks]

2. A company is concerned about the level of information they have available online.

- (a) Is there any type of penetration testing that can be considered realistically possible with a duration of 2 weeks? Your answer should not exceed 200 words.

[3 marks]

- (b) From Figure 1 & 2 (next page) can you extract at least three pieces per figure of vital information and explain how an attacker might use them in a malicious way (3 pieces of information for each image; 6 in total)? Your answer should not exceed 700 words.

[12 marks]

HOSTS:		
45.33.32.156	Linux	Apache/2.4.7 Ubuntu
74.207.244.221	Linux	Apache/2.2.14 Ubuntu
Application Server:		
Technology		Description
Apache 🔗		Web server software
Client-Side		
Includes all the main technologies that run on the browser (such as JavaScript and Adobe Flash).		
Technology		Description
JavaScript 🔗		Widely-supported programming language commonly used to power client-side dynamic content on websites

Figure 1: "Example a"

IN	AAAA	2600:3c01::f03c:91ff:fe18:bb2f				
IN	A	45.33.32.156				
IN	AAAA	2600:3c01::f03c:91ff:fe98:ff4e				
IN	TXT	v=spf1 a mx ptr ip4:45.33.49.119 ip4:173.255.243.189 ip4:192.81.131.254 ip6:2600:3c01::f03c:91ff:fe98:ff4e ip6:2600:3c01::f03c:91ff:fe70:d085 include:_spf.google.com ~all				
IN	MX	<table><tr><td>preference:</td><td>10</td></tr><tr><td>exchange:</td><td>aspmx2.googlemail.com</td></tr></table>	preference:	10	exchange:	aspmx2.googlemail.com
preference:	10					
exchange:	aspmx2.googlemail.com					
IN	MX	<table><tr><td>preference:</td><td>5</td></tr><tr><td>exchange:</td><td>alt2.aspmx.l.google.com</td></tr></table>	preference:	5	exchange:	alt2.aspmx.l.google.com
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exchange:	alt2.aspmx.l.google.com					
IN	MX	<table><tr><td>preference:</td><td>10</td></tr><tr><td>exchange:</td><td>aspmx3.googlemail.com</td></tr></table>	preference:	10	exchange:	aspmx3.googlemail.com
preference:	10					
exchange:	aspmx3.googlemail.com					

Figure 2: "Example b"

- (c) Can you mention some of the tools that could produce similar results as figure 1 & 2 (figures seen in previous question)? Your answer should not exceed 150 words. [2 marks]

(d) Figure 3 (below) shows an output from a different tool. Describe how this output differs from that shown in figure 1 & 2 (figures seen in question 2b). Your answer should not exceed 200 words.

[3 marks]

```
Host is up (0.17s latency).
Not shown: 498 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.10 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
Aggressive OS guesses: Linux 3.13 or 4.2 (96%), Linux 3.10 - 4.8 (95%), Linux 3.2 - 4.8 (93%), Linux 3.13 (92%), Linux 4.4 (92%), Asus RT-AC66U WAP (92%),
Linux 3.10 (92%), Linux 3.11 - 3.12 (92%), Linux 3.18 (92%), Linux 3.2 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 18 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 3: "Scan example"

(e) If an attacker had all this information, how could they use it for exploitation? Please describe two scenarios. Your answer should not exceed 250 words

[5 marks]

[Total: 25 marks]

3. A computer science student regularly works from home, using their computer to access remote machines on campus to run code, download papers from web archives, send email and upload files to the university fileserver.

(a) What services and ports would you expect the student to use (you need to mention at least five)? Your answer should not exceed 650 words.

[11 marks]

(b) Describe two potential attack scenarios the student may be vulnerable to, based upon the information gathered in Q3a including. Your answer should not exceed 300 words.

[4 marks]

[Total: 15 marks]